Assignment 6 杜奕旋 12290095

1. (a)

	SS	df	MS	F
Treatment	22385775.639642	2	11192887.819821	338.384375
Error	118348579728.59166	48201	2455313.78454	NaN
Total	118370965504.231308	48203	None	NaN

(b) H_0: $\mu_{cloud} = \mu_{clear} = \mu_{snow}$

H_1: At least one μ_i is different

Treatment: $df = m - 1 = 2$

Error: $df = n - m = 48201$

$$F = \frac{MS(T)}{MS(E)} = 338.384375$$

$$F_{0.05}(2, 48201) = 2.99592 < F$$

$\therefore$ H_0 is rejected, which means at least one μ_i is different

2. (a)

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Brand	3	1220	406.7	0.613	0.620
Surface	2	1848	924.0	1.393	0.286
Brand:Surface	6	8477	1412.9	2.129	0.125
Residuals	12	7963	663.5		
Total	23				

(b) H_0: $\mu_{Brand1} = \mu_{Brand2} = \mu_{Brand3} = \mu_{Brand4}$

H_1: At least one μ_i is different.

$\because Pr(>F) = 0.620 > 0.05$

$\therefore$ Fail to reject H_0, so there is no significant effect of the brand.

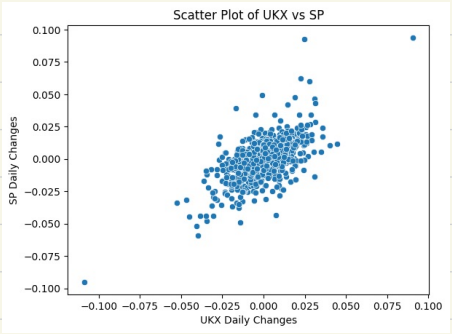
(c) H_0: $\mu_{surface1} = \mu_{surface2} = \mu_{surface3}$

H_1: At least one μ_i is different

$\because Pr(>F) = 0.286 > 0.05$ $\therefore$ Fail to reject H_0. There is no significant column effect.

(d) H_0: There is no significant interaction effect
H_1: There is significant interaction effect.
∵ $Pr(>F) = 0.125 > 0.05$ ∴ Fail to reject H_0.
∴ There is no significant interaction effect.

3. (a)



- The two indexes seem to change in the same direction.
- Maybe reasonable to apply the simple linear regression

(b)

OLS Regression Results

Dep. Variable:	SP	R-squared:	0.375
Model:	OLS	Adj. R-squared:	0.375
Method:	Least Squares	F-statistic:	1065.
Date:	Sun, 10 Dec 2023	Prob (F-statistic):	1.94e-183
Time:	14:16:45	Log-Likelihood:	5938.2
No. Observations:	1778	AIC:	-1.187e+04
Df Residuals:	1776	BIC:	-1.186e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
const	0.0005	0.000	2.247	0.025	5.81e-05	0.001
UKX	0.6437	0.020	32.637	0.000	0.605	0.682

Omnibus:	306.343	Durbin-Watson:	2.343
Prob(Omnibus):	0.000	Jarque-Bera (JB):	4545.311
Skew:	0.322	Prob(JB):	0.00
Kurtosis:	10.806	Cond. No.	96.9

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

$SP = \underset{(0.020)}{0.0005} + 0.6437\,UKX$

(c) F-test = 1065.,
$R^2 = 0.375$

(d) The 95% CI for intercept: [5.81e-05, 0.001]
for slope : [0.605, 0.682]

(e) $H_0: \beta_1 = 0$

$H_1: \beta_2 \neq 0$

Test for Constraints

	coef	std err	t	P>|t|	[0.025	0.975]
c0	0.6437	0.020	32.637	0.000	0.605	0.682

$t_{\frac{0.05}{2}}(1776) < t = 32.637$ $\therefore$ reject H_0

$\therefore$ the slope term is significantly different from zero.

(f) $H_0: \beta_1 \leq 0.5$

$H_1: \beta_1 > 0.5$

Test for Constraints

	coef	std err	t	P>|t|	[0.025	0.975]
c0	0.6437	0.020	7.286	0.000	0.605	0.682

$t_{0.05}(1776) < t = 7.286$ $\therefore$ reject H_0

$\therefore$ The slope is significantly larger than 0.5

(g)

95% Confidence Intervals:

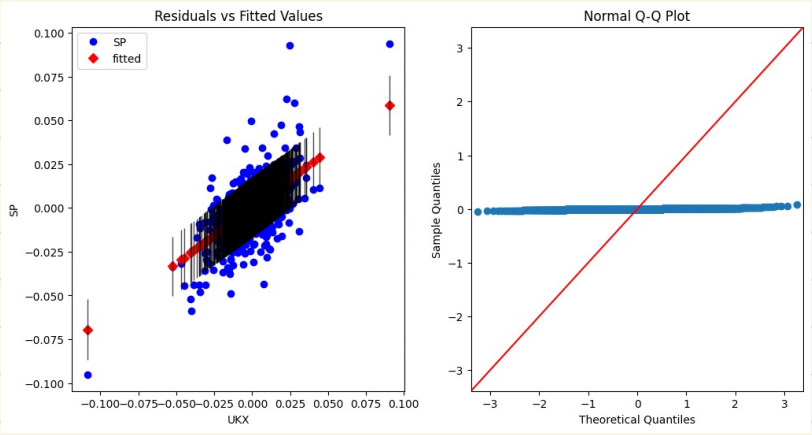
	mean_ci_lower	mean_ci_upper
0	0.605012	0.682377

95% Prediction Intervals:

	obs_ci_lower	obs_ci_upper
0	0.601509	0.685879

This means that if the UKX daily changes are 0, we are 95% confident that the true mean change in SP daily changes falls within [0.605012, 0.682377]; similarly, we are 95% confident that the true change in a new, individual SP daily changes fall within [0.601509, 0.685879]

(h)



The model assumptions seem to be satisfied
The Q-Q plot is likely a straight line.

4. $k_i = \dfrac{x_i - \bar{x}}{\sum_{i=1}^{n}(x_i-\bar{x})^2}$

(a) $\sum_{i=1}^{n} k_i = \dfrac{(x_1-\bar{x})+(x_2-\bar{x})+\cdots+(x_n-\bar{x})}{\sum_{i=1}^{n}(x_i-\bar{x})^2} = \dfrac{n\bar{x}-n\bar{x}}{\sum_{i=1}^{n}(x_i-\bar{x})^2} = 0$

$\therefore k_1+\cdots+k_n = 0$

(b) $k_1x_1+\cdots+k_nx_n = \dfrac{(x_1-\bar{x})x_1+(x_2-\bar{x})x_2+\cdots+(x_n-\bar{x})x_n}{\sum_{i=1}^{n}(x_i-\bar{x})^2} = \dfrac{\sum_{i=1}^{n}x_i^2-n\bar{x}^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2}$

$\sum_{i=1}^{n}(x_i-\bar{x})^2 = \sum_{i=1}^{n}(x_i^2-2\bar{x}x_i+\bar{x}^2) = \sum_{i=1}^{n}x_i^2-2n\bar{x}^2+n\bar{x}^2 = \sum_{i=1}^{n}x_i^2-n\bar{x}^2$

$\therefore k_1x_1+\cdots+k_nx_n = 1$

(c) $k_i^2 = \dfrac{(x_i-\bar{x})^2}{\left[\sum_{i=1}^{n}(x_i-\bar{x})^2\right]^2}$

$\sum k_i^2 = \dfrac{(x_1-\bar{x})^2+(x_2-\bar{x})^2+\cdots+(x_n-\bar{x})^2}{\left[\sum_{i=1}^{n}(x_i-\bar{x})^2\right]^2} = \dfrac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{\left[\sum_{i=1}^{n}(x_i-\bar{x})^2\right]^2} = \dfrac{1}{\sum_{i=1}^{n}(x_i-\bar{x})^2}$

$\therefore k_1^2+\cdots+k_n^2 = \dfrac{1}{\sum_{i=1}^{n}(x_i-\bar{x})^2}$

Q2 R code

```
> # 创建包含给定数据的数据框
> data <- data.frame(
+   Brand = rep(c("Brand1", "Brand2", "Brand3", "Brand4"), each = 2 * 3),  # 4$
+   Surface = rep(rep(1:3, each = 2), times = 4),  # 写作表面1到3，每个Surface $
+   Lifetime = c(
+     709, 659, 713, 726, 660, 645,  # Brand1对应的测量值
+     668, 685, 722, 740, 692, 720,  # Brand2对应的测量值
+     698, 685, 666, 684, 678, 750,  # Brand3对应的测量值
+     698, 650, 704, 666, 686, 733   # Brand4对应的测量值
+   )
+ )
>
> # 将 "Surface" 变量转换为因子，以确保正确的水平（levels）
> data$Surface <- factor(data$Surface)
>
> # 进行两因素方差分析
> model <- aov(Lifetime ~ Brand * Surface, data = data)
>
> # 显示方差分析表
> summary(model)
```